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United States Patent	12385491
Kind Code	B2
Date of Patent	August 12, 2025
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Direct drive refrigerant screw compressor with refrigerant lubricated rotors

Abstract

Disclosed is a direct-drive refrigerant screw compressor, having: a housing; a compression chamber in the housing; a pair of rotors, each rotor of the pair of rotors being rotationally disposed in the compression chamber and including an outer surface with a screw-gear profile; a fluid being disposed in the compression chamber, the fluid consisting of a working fluid for providing lubrication to each rotor; a first port extending through the housing and configured for directing the fluid toward the compression chamber; and when the compressor is activated, each rotor rotates and the fluid is distributed about each rotor to lubricate each rotor.

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Appl. No.: 18/524682

Filed: November 30, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240102472 A1	Mar. 28, 2024

Related U.S. Application Data

division parent-doc US 16973724 US 11898561 WO PCT/US2020/033585 20200519 child-doc US 18524682

Publication Classification

Int. Cl.: **F04C29/02** (20060101); **F04C18/16** (20060101); **F25B1/047** (20060101)

U.S. Cl.:

CPC **F04C29/02** (20130101); **F04C18/16** (20130101); **F04C29/023** (20130101); **F04C29/028** (20130101); F04C29/026 (20130101); F04C2210/10 (20130101); F04C2210/14 (20130101); F04C2240/20 (20130101); F04C2240/50 (20130101); F25B1/047 (20130101)

Field of Classification Search

CPC: F04C (18/16); F04C (29/02); F04C (29/023); F04C (29/026); F04C (29/028); F04C (2210/10); F04C (2210/14); F04C (2240/20); F04C (2240/50); F25B (1/047)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 16/973,724 filed Dec. 9, 2020, which is a 371 of PCT Application No. PCT/US2020/033585 filed May 19, 2020, and claims the benefit of U.S. Provisional Patent Application No. 62/850,296, filed on May 20, 2019, each of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) The disclosure relates generally to compressor systems and, more specifically, to a direct drive refrigerant screw compressor using refrigerant lubrication of one or more components thereof.

(2) Refrigeration systems are utilized in many applications to condition an environment. The cooling or heating load of the environment may vary with ambient conditions, occupancy level, other changes in sensible and latent load demands, and with temperature and/or humidity changes.

(3) Refrigeration systems typically include a compressor to deliver compressed refrigerant to a condenser. From the condenser, the refrigerant travels to an expansion valve and then to an evaporator. From the evaporator, the refrigerant returns to the compressor to be compressed.

(4) A direct drive screw compressor in an HVAC chiller application has a driving (male) rotor and a driven (female) rotor. An electric motor drives the driving rotor to rotate. The driving rotor then drives the driven rotor by way of meshing. The meshing process requires direct contact of the rotors at contact locations. Lubrication is necessary to protect both rotors and decrease the friction during operation. In addition, the rotors in a screw compressor in HVAC chiller applications are supported by rolling element bearings. These bearings may be lubricated using oil because of a high viscosity requirement of bearing lubricant. After passing through the bearings, oil is mixed with refrigerant in the compression process to be carried out of the compressor.

BRIEF DESCRIPTION

(5) Disclosed is a direct-drive refrigerant screw compressor, comprising: a housing; a compression chamber in the housing; a pair of rotors, each rotor of the pair of rotors being rotationally disposed in the compression chamber and including an outer surface with a screw-gear profile; a fluid being disposed in the compression chamber, the fluid consisting of a working fluid for providing lubrication to each rotor; a first port extending through the housing and configured for directing the

fluid toward the compression chamber; and when the compressor is activated, each rotor rotates and the fluid is distributed about each rotor to lubricate each rotor.

(6) In addition to one or more of the above features, or as an alternate, the first port includes a flow control orifice.

(7) In addition to one or more of the above features, or as an alternate, the first port extends directly into the compression chamber.

(8) In addition to one or more of the above features, or as an alternate, the first port is fluidly connected to a passage in one rotor of the pair of rotors that directs the fluid to the compression chamber.

(9) In addition to one or more of the above features, or as an alternate, the passage extends between an axial aft port in the one rotor and the outer surface of the one rotor.

(10) In addition to one or more of the above features, or as an alternate, the passage includes an axial segment forming a blind hole and a radial segment fluidly connected between the axial segment and a surface port on the outer surface of the one rotor.

(11) In addition to one or more of the above features, or as an alternate, the passage includes a plurality of the radial segments fluidly connected to a respective plurality of the surface ports on the outer surface of the one rotor.

(12) In addition to one or more of the above features, or as an alternate, the plurality of the surface ports are staggered at regular intervals along the outer surface of the one rotor.

(13) In addition to one or more of the above features, or as an alternate, the plurality of the radial segments each include opposing radial portions extending to a respective plurality of the surface ports on the outer surface of the one rotor.

(14) Further disclosed is a refrigerant system including: a condenser; a compressor having one or more of the above disclosed features; and a conduit fluidly connecting the condenser and the first port of the compressor, and configured to transport the fluid to the compressor to provide the working fluid to each rotor.

(15) Further disclosed is a method of directing fluid in a direct drive screw compressor, comprising: receiving fluid at a first port of a housing of the compressor, wherein the fluid consists of a working fluid for providing lubrication to each rotor of a pair of rotors in the compressor; and directing the fluid from the first port to a compression chamber in the compressor; and when the compressor is activated, each rotor rotates and the fluid is distributed about each rotor to lubricate each rotor.

(16) In addition to one or more of the above features, or as an alternate, the method includes controlling flow through the first port with a flow control orifice.

(17) In addition to one or more of the above features, or as an alternate, directing the fluid to the compression chamber includes: injecting the fluid from the first port directly into the compression chamber.

(18) In addition to one or more of the above features, or as an alternate, directing the fluid to the compression chamber includes: injecting the fluid from the first port, through a passage in one rotor of the pair of rotors, whereby the fluid is injected into the compression chamber.

(19) In addition to one or more of the above features, or as an alternate, injecting the fluid through the passage includes: directing the fluid from the first port into an axial aft port in the passage and out an outer surface of the one rotor.

(20) In addition to one or more of the above features, or as an alternate, directing the fluid through the passage further includes: directing the fluid through an axial segment forming a blind hole in the one rotor and a radial segment fluidly connected between the axial segment and a first surface port on the outer surface of the one rotor.

(21) In addition to one or more of the above features, or as an alternate, directing the fluid through the passage further includes: directing the fluid through a plurality of the radial segments fluidly connected to a respective plurality of the surface ports on the outer surface of the one rotor.

(22) In addition to one or more of the above features, or as an alternate, directing the fluid through the passage further includes: directing the fluid through opposing radial portions of each of the plurality of the radial segments, the opposing radial portions extending to a respective plurality of the surface ports on the outer surface of the one rotor.

(23) In addition to one or more of the above features, or as an alternate, receiving the fluid at the first port from a condenser in a refrigerant system in which the compressor is integrated, to provide the working fluid to each rotor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

(2) FIG. 1 is a refrigerant system in which features of the disclosed embodiments may be utilized;

(3) FIG. 2 is a refrigerant system according to a disclosed embodiment;

(4) FIG. 3 is a direct-drive screw compressor according to one embodiment;

(5) FIG. 4 is a direct-drive screw compressor according to one embodiment;

(6) FIG. 5 is a direct-drive screw compressor according to one embodiment;

(7) FIG. 6 is a method of transporting refrigerant as a lubricant with the compressor of FIG. 4;

(8) FIG. 7 is a method of transporting refrigerant as a lubricant with the compressor of FIG. 5;

(9) FIG. 8 is a direct-drive screw compressor according to one embodiment; and

(10) FIG. 9 is a method of transporting refrigerant as a lubricant with the compressor of FIG. 8.

DETAILED DESCRIPTION

(11) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

(12) Described herein are systems and methods for lubricating components of a compressor in a refrigeration system. FIG. 1 illustrates a refrigeration system 10 that is an oil lubricated system.

The system 10 includes a condenser 15 that receives a high pressure gaseous form of the working fluid, ejects heat from the working fluid, for example to the environment, and outputs a high pressure liquid form of the working fluid. Downstream of the condenser 15 is an expansion valve 20 that receives the high pressure liquid form of the working fluid and outputs a low pressure liquid form of the working fluid. Downstream of the expansion valve 20 is an evaporator 25 that receives the low pressure liquid form of the working fluid, transfers heat to the working fluid, thereby conditioning warm air, and outputs a low pressure gaseous form of the working fluid. Downstream of the evaporator 25 is a compressor 30 that receives the low pressure gaseous form of the working fluid and outputs a high pressure gaseous form of the working fluid.

(13) The compressor 30 may be a screw compressor that includes suction bearings 35, discharge bearings 40, and a set of rotors 45 therebetween. Both sets of bearings 35, 40 and the rotors 45 require some form of lubrication. Lubricating oil is provided by an oil separator 50. The oil separator 50 transfers oil to an oil filter 55. The oil filter 55 transfers oil a first portion of oil 60 to one orifice 71, e.g. in the compressor housing, fluidly connected to the suction bearings 35. A second portion of oil 65 is distributed in parallel to one orifice 70, e.g., in the compressor housing, fluidly connected to the rotors 45 and another orifice 75, e.g., in the compressor housing, fluidly connected to the discharged bearings 40. The oil then mixes with the working fluid in the compressor 30.

(14) Output from the compressor 30 is directed to the oil separator 50. The oil separator 50 separates the output from the compressor into a first portion 80 that is the working fluid directed to the condenser 15. The second portion 85 is the lubricant directed to the filter 55. Unless otherwise indicated herein, for each embodiment all flows between the system components that are separately

referred to are fluidly transferred in respective conduit lines. It is to be appreciated that fluid branches that are branched upstream or downstream of the orifices **70, 75** in the housing of the compressor **30** may be branched in conduit exterior to the housing of the compressor **30**.

(15) Viscosity of oil lubricant may be reduced when mixed with the working fluid. Both bearing load carrying capacity and oil sealing characteristics are dependent upon the oil viscosity. As such, due to lower viscosity, moving components, such as bearings and rotors, in some systems may experience increased wear during operation. In addition, separating lubricating oil from refrigerant requires the use and maintenance of additional equipment such as the oil separator and related filter. In addition, because the oil separation process cannot completely remove the oil from refrigerant, excessive oil may decrease heat transfer efficiency in the system and lower the overall system capacity. Oil may be saturated with refrigerant in the separator. The separation process is often unable to adequately lower the refrigerant content in the oil.

(16) In view of the above challenges FIGS. 2-7 disclose embodiments in which an oil separator and oil filter may be avoided. More specifically, turning to FIG. 2, disclosed is a refrigerant system **100** (a chiller) applicable to each of the embodiments disclosed herein. The system **100** includes a condenser **110**, an expansion valve **112**, an evaporator **114**, and a dual rotor refrigerant screw compressor **115** (compressor **115**), which is a direct drive compressor. The compressor **115** includes two screw rotors **150**. The rotors **150** are configured in the compressor **115** with a suction side **140a** and discharge side **140b** (illustrated schematically in FIG. 2). The compressor **115** includes bearing packs **190** including a suction side bearing pack **190a** and a discharge side bearing pack **190b**. The suction side bearing pack **190a** may be referred to herein as a forward bearing pack and the discharge side bearing pack **190b** may be referred to herein as an aft bearing pack.

(17) The condenser feeds first portion **116** of a working fluid to the expansion valve **112** and, in parallel, a second portion **120** of the working fluid **120** to the compressor **115**. The working fluid consists of refrigerant from a condenser conduit **125** to the compressor **115** for providing lubrication to components of the compressor **115** as described below.

(18) The second portion **120** of the working fluid is distributed in parallel to a first branch **121** and a second branch **122**. The first branch **121** is distributed in parallel to a third branch **123** and a fourth branch **124**. The third branch **123** delivers the working fluid through one or more orifices **126**, e.g. in the compressor housing **130**, to the suction side bearing pack **190a**. The fourth branch **124** delivers the working fluid through another one or more orifices **127**, e.g. in the compressor housing **130**, to the rotors **150**. The second branch **122** delivers the working fluid to a further one or more orifices **128**, e.g. in the compressor housing **130**, to the branch side bearing pack **190b**.

(19) From the suction side bearing pack **190a**, the working fluid flows directly into the rotors **150** with the working fluid from the evaporator **114**. This may occur within the compressor housing **130**. From the discharge side bearing pack **190b** the working fluid flows to the evaporator **114** to mix with fluid therein and then be redirected to the rotors **150** of the compressor **115**. This may occur by the working fluid exiting the compressor housing **130** from the discharged side bearings **190b** and being directed thereafter to the evaporator **114**. Unless otherwise indicated herein, for each embodiment all flows between the system components that are separately referred to are fluidly transferred in respective conduit lines. It is to be appreciated that fluid branches that are branched upstream or downstream of the orifices **126, 127, 128** in the compressor housing **130** may be branched in conduit exterior to the compressor housing **130**.

(20) The features of the compressor are illustrated more specifically, for example, in FIGS. 3-5. Turning now to FIG. 3, the compressor **115** includes the housing **130**. A compression chamber **140** is disposed in the housing **130**. The compression chamber **140** has a forward end **140a** and an aft end **140b** which are respective suction and discharge sides of the compression chamber **140**. For simplicity, inlet and outlet ports in the housing **130** for fluidly communicating working fluid **120** in the refrigeration system **100** are not illustrated in FIG. 3.

(21) The compressor **115** includes the plurality of rotors generally referred to as **150**, including the

first rotor **150a** and the second rotor **150b**, rotationally disposed in the compression chamber **140**. Each rotor **150** includes an outer surface **160** with a screw-gear profile, for example, having an alternating plurality of peaks **160a** and plurality of troughs **160b**, for example, in cross sectional view. The plurality of rotors **150** intermesh and form compression volumes within the compression chamber **140**. The first rotor **150a** is a driven rotor and the second rotor **150b** is a drive rotor, driven by a motor **180**.

(22) For each rotor **150**, the compressor **115** includes the plurality of bearing packs generally referred to as **190** including the forward bearing pack generally referred to as **190a** and the aft bearing pack generally referred to as **190b**. For each rotor **150**, the plurality of bearing packs **190** may disposed within a respective plurality of bearing chambers generally referred to as **200**. The bearing chambers **200** may be structural portions of the housing **130** in or proximate the compression chamber **140** configured to securely position the respective bearing packs **190**. The bearing chambers **200** may including a forward bearing chamber generally referred to as **200a** and an aft bearing chamber generally referred to as **200b**. The bearing chambers **200** may be fluidly connected with each other through the compression chamber **140**.

(23) Turning now to FIG. **4**, an embodiment of the refrigeration system **100** is illustrated. The embodiment of FIG. **4** includes all of the features illustrated in the system **100** illustrated in FIG. **3**. In FIG. **4**, the fluid **120** is disposed within the compression chamber **140**. A first port **220** extends through the housing **130** for directing fluid toward the compression chamber **140**. The first port **220** is connected by the condenser conduit **125** to the condenser **110**. According to an embodiment, the first port **220** includes a flow control orifice **230**. This may be used to reduce a flow volume or rate from the condenser **110** as may be needed.

(24) In FIG. **4**, the first port **220** extends directly into the compression chamber **140**. Within the compression chamber **140**, the first port **220** delivers working fluid **120** between the two rotors **150** so that the working fluid **120** flows to meshing points between the two rotors **150**. In one embodiment, the first port **220** is proximate one rotor **150** (the second rotor **150b**) of the compressor **115** and distal the other rotor **150** (the first rotor **150a**). Identifying the one rotor **150** as the second rotor **150b** and the other rotor **150** as the first rotor **150a** in the embodiment in FIG. **4** is for example only and not intended on limiting the scope of the embodiments. Rotation of the rotors **150** distributes the fluid **120** about the rotors **150**.

(25) Turning now to FIG. **5**, an embodiment of the refrigeration system **100** is illustrated. The embodiment of FIG. **5** includes all of the features illustrated in the system **100** illustrated in FIG. **3**. In FIG. **5**, the fluid **120** is disposed within the compression chamber **140**. A first port **220**, configured differently than the first port **220** in the embodiment of FIG. **4**, extends through the housing **130**. In FIG. **5**, the first port **220** fluidly connects with a passage **260** within one rotor **150** (the first rotor **150a**) for directing fluid toward the compression chamber **140**. Identifying the one rotor **150** as the first rotor **150a**, and thus the other rotor **150** as the second rotor **150a**, in the embodiment in FIG. **5** is for example only and not intended on limiting the scope of the embodiments. The first port **220** is connected by the condenser conduit **125** to the condenser **110**. According to an embodiment, the passage **260** includes a flow control orifice **230**, which may be the same as the above introduced flow control orifice **230**. This may be used to reduce a flow volume or rate from the condenser **110** as may be needed.

(26) The passage **260** may be an internal passage in the one rotor **150**. The passage **260** may be fluidly connected between an axial aft port **265** in the one rotor **150** and the outer surface **160** of the one rotor **150**. The aft port **265** may be in the respective aft bearing chamber **200b**, though this placement is not intended to be limiting.

(27) The passage **260** may include an axial segment **270** forming a blind hole in the one rotor **150** and a radial segment generally referred to as **280** fluidly connected between the axial segment **270** and a surface port generally referred to as **290** on the outer surface **160** of the one rotor **150**. In one embodiment, the passage **260** may include a plurality of the radial segments **280** fluidly connected

to a respective plurality of the surface ports **290** on the outer surface **160** of the one rotor **150**. This configuration may provide a greater distribution of the fluid **120** about each rotor **150** as compared with, for example, a single fluid **120** port.

(28) In one embodiment, the plurality of the surface ports **290** may be staggered at regular intervals along the outer surface **160**, for example, at or proximate the plurality of alternating peaks **160a** or troughs **160b**. This configuration may provide an even distribution of fluid **120** around the outer surface **160** of the each rotor **150**. In one embodiment the plurality of the radial segments **280** may each include a plurality of opposing radial portions **280a**, **280b** extending to a respective plurality of the radial ports **290a**, **290b** on the outer surface **160** of the one rotor **150**. This configuration may provide an ability to quickly distribute fluid **120** around the outer surface **160** of the rotors **150**.

(29) Turning to FIG. 5, a method is disclosed of directing fluid **120** in the compressor **115** for the embodiment illustrated in FIG. 3. The method includes block **510** of receiving the fluid **120** at the first port **220** of the housing **130**. In an embodiment, block **510** further includes controlling flow in the first port **220** through a flow control orifice **230** (which may be the same as orifice **127** in FIG. 2). The method further includes block **520** of directing the fluid **120** in the compressor **115**, from the first port **220**, to the compression chamber **140**. According to an embodiment, block **520** further includes injecting the fluid **120** from the first port **220** directly into the compression chamber **140** proximate one rotor **150** and distal the other rotor **150**. At block **530** the compressor is activated to distribute the fluid about the rotors **150**.

(30) Turning to FIG. 7, a method is disclosed of directing fluid **120** in the compressor **115** for the embodiment illustrated in FIG. 5. Similar to the method in FIG. 6, the method of FIG. 7 includes block **610** of receiving the fluid **120** at the first port **220** of the housing **130**. The method of FIG. 7 includes block **620** of directing the fluid **120**, from the first port **220**, to the compression chamber **140**. In an embodiment, block **620** further includes controlling flow in the passage **260** through a flow control orifice **230**. In an embodiment, block **620** further includes injecting the fluid **120** through the first port **220**, through a passage **260** in one rotor **150**, and into the compression chamber **140**. Then, at block **630** the compressor is activated to distribute the fluid about the rotors **150**.

(31) Thus, in the above disclosed embodiments, the working fluid **120** is drawn from a chiller condenser and used to provide lubrication to the compressor and more specifically to the screw rotors. The liquid can be injected direct from port(s) on the housing close to the rotor meshing locations or through a passage inside the driving rotor. The liquid flow can be adjusted by using flow restriction devices, such as a flow control orifice. The embodiments enable the utilization of pure refrigerant as the working fluid **120** in the components of the system **100**, including the condenser **110**, evaporator **114**, etc.

(32) Turning now to FIG. 8 a further embodiment of a refrigerant system **100** is illustrated. The embodiment of FIG. 8 includes all of the features illustrated in the system **100** illustrated in FIG. 3. In FIG. 8, the fluid **120** is disposed within each of the plurality of bearing chambers **200** for providing lubrication to the plurality of bearing packs **190**, thus providing pure refrigerant lubricated (PRL) bearings. A plurality of bearing lubrication ports generally referred to as **300** extend through the housing **130** and into each of the plurality of bearing chambers **200**.

(33) In addition, a suction side (upstream) lubrication port **300a** includes a suction side (upstream) flow control orifice **301a** (which may be the same as orifice **126** in FIG. 2). A discharge side (downstream) lubrication port **300b** includes a discharge side (downstream) flow control orifice **301b** (which may be the same as orifice **128** in FIG. 2).

(34) The condenser conduit **125** fluidly connects the condenser **110** to the plurality of bearing lubrication ports **300**. From this configuration, the plurality of bearing lubrication ports **300** are configured for injecting the fluid **120** into each of the plurality of bearing chambers **200** when the compressor **115** is running, to thereby provide lubrication to the plurality of bearing packs **190**. In one embodiment the plurality of bearing lubrication ports **300** include a respective plurality flow

control orifices **230** to reduce a flow volume or rate from the condenser **110** as may be needed.

(35) In one embodiment, the condenser conduit **125** includes a forward branch **310a** and an aft branch **310b** for injecting in parallel the fluid **125** to each forward bearing chamber **200a** and each aft bearing chamber **200b** in the compressor. Each branch **310a**, **310b** includes a plurality of sub-branches **320a**, **320b**, generally referred to as **320**, for injecting in parallel the fluid to the bearing chambers **200** on each branch **310a**, **310b**. This configuration enables the condenser **110** to feed the fluid **120** to the compressor **115** from the single condenser conduit **125**.

(36) As further illustrated in FIG. **8**, for each rotor **150** the compressor **115** includes a lubricant drain port generally referred to as **360** fluidly connected to the evaporator by an evaporator conduit **370**. The lubricant drain port **360** is for draining the fluid **120** from the plurality of bearing chambers **200** of the respective rotor **150** when the compressor **115** is running. In one embodiment, each lubricant drain port **360** extends into the respective aft bearing chamber **200b** and is fluidly connected to the respective forward bearing chamber **200a** through the respective aft bearing chamber **200b**.

(37) As illustrated in FIG. **9**, a further method is disclosed of directing fluid **120** in the compressor **115** in the refrigerant system **100**. The method includes block **710** of receiving the fluid **120** from the compressor **115** in the refrigerant system **100**, through a condenser conduit **125**, at the plurality of bearing lubrication ports **300**. The method includes block **720** of directing the fluid **120** through the plurality of bearing lubrication ports **300** to the plurality of bearing chambers **200**. From this configuration the fluid **120** is injected, when the compressor **115** is running, to the plurality of bearing packs **190** in the respective plurality of bearing chambers **200**. According to an embodiment, box **710** may further include controlling flow through the plurality of bearing lubrication ports **300** with a respective plurality of flow control orifices **230**. Then, at block **725** the compressor is activated to distribute the fluid about the rotors **150**. That is, the fluid **130** is inject to one side of the bearing packs **190** and is flow through the bearing packs **190** to lubricate each of the bearing packs **190**.

(38) According to an embodiment, for each rotor **150**, the method includes block **730** of draining the fluid **120** through the lubricant drain port **360** from the plurality of bearing chambers **200** when the compressor **115** is running. According to an embodiment, for each rotor **150** block **730** further includes draining the fluid **120** from the plurality of chambers **20** through the aft bearing chamber **200**, into the evaporator conduit **370**, and to the evaporator **114** in the refrigerant system **100**.

(39) With the above disclosed embodiments, for example in FIGS. **3**, **8** and **9**, pure refrigerant lubricated (PRL) bearings are used in a screw compressor to support the loads on the rotors. The PRL bearings operate with a relatively low viscosity lubricant, such as liquid refrigerant as the working fluid. The liquid refrigerant as the working fluid is drawn from the chiller condenser and injected directly to each individual bearings or pack of bearings. The liquid flow can be adjusted by using flow restriction devices, such as an orifice.

(40) With the above disclosed embodiments, oil separation equipment on a chiller is no longer necessary. This configuration reduces the complexity of the chiller system. The chiller cost will be therefore reduced. The chiller heat transfer efficiency will therefore increase.

(41) Accordingly, as indicated above, there are two kinds of fluids in a typical system: oil and a working fluid. Oil is typically used for lubricating bearings and rotors and for sealing. The working fluid, such as refrigerant, is typically used to transmit heat. According to the disclosed embodiments, the working fluid, instead of oil, is used for lubricating bearings and rotors.

(42) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.

(43) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used

in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(44) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A direct-drive refrigerant screw compressor, comprising: a housing; a compression chamber in the housing; a pair of rotors, each rotor of the pair of rotors being rotationally disposed in the compression chamber and including an outer surface with a screw-gear profile; a fluid being disposed in the compression chamber, the fluid consisting of a working fluid for providing lubrication to each rotor, wherein the working fluid is refrigerant delivered from a condenser conduit to the compressor for providing lubrication to compressor; a first port extending through the housing and configured for directing the fluid toward the compression chamber; wherein: when the compressor is activated, each rotor rotates and the fluid is distributed about each rotor to lubricate each rotors for each of the rotors, the compressor includes a plurality of bearing packs disposed within a respective plurality of bearing chambers, the plurality of bearing chambers being structural portions of the housing that are in or proximate the compression chamber, the respective plurality of bearing chambers being configured to securely position the respective bearing packs, the plurality of bearing chambers including a forward bearing chamber and an aft bearing chamber for each of the rotors, the forward and aft bearing chambers being fluidly connected with each other through the compression chamber of each of the rotors, wherein: the fluid is directed to the forward and aft bearing chambers for each of the rotors within the compressor that are fluidly coupled to the compression chamber; and upon activation of the compressor, the fluid is injected to one side of the plurality of bearing chambers and flows through the plurality of bearing chambers to lubricate each of the plurality of bearing packs and is distributed about the rotors; and the condenser conduit includes a forward branch and an aft branch for injecting in parallel the working fluid to the forward bearing chamber and the aft bearing chamber for each of the rotors of the compressor, each of the forward and aft branches of the condenser conduit includes a plurality of sub-branches for injecting in parallel the working fluid to the plurality of bearing chambers on each of the forward and aft branches of the condenser conduit.
2. The compressor of claim 1, wherein: the first port includes a flow control orifice.
3. The compressor of claim 1, wherein: the first port extends directly into the compression chamber.
4. The compressor of claim 1, wherein: the first port is fluidly connected to a passage in one rotor of the pair of rotors that directs the fluid to the compression chamber.
5. The compressor of claim 4, wherein: the passage extends between an axial aft port in the one rotor and the outer surface of the one rotor.
6. The compressor of claim 5, wherein: the passage includes an axial segment forming a blind hole and a radial segment fluidly connected between the axial segment and a surface port on the outer surface of the one rotor.
7. The compressor of claim 6, wherein: the passage includes a plurality of the radial segments fluidly connected to a respective plurality of the surface ports on the outer surface of the one rotor.

8. The compressor of claim 7, wherein: the plurality of the surface ports are staggered at regular intervals along the outer surface of the one rotor.

9. The compressor of claim 8, wherein: the plurality of the radial segments each include opposing radial portions extending to a respective plurality of the surface ports on the outer surface of the one rotor.

10. A refrigerant system including: a condenser; the compressor of claim 1; and the condenser conduit fluidly connecting the condenser and the first port of the compressor, and configured to transport the fluid to the compressor to provide the working fluid to each rotor.

11. A method of directing fluid in a direct drive screw compressor, comprising: receiving fluid at a first port of a housing of the compressor, wherein the fluid consists of a working fluid for providing lubrication to each rotor of a pair of rotors in the compressor; and directing the fluid from the first port to a compression chamber in the compressor; and when the compressor is activated, each rotor rotates and the fluid is distributed about each rotor to lubricate each rotor, wherein the working fluid is refrigerant delivered from a condenser conduit to the compressor for providing lubrication to compressor; wherein: for each of the rotors, the compressor includes a plurality of bearing packs disposed within a respective plurality of bearing chambers, the plurality of bearing chambers being structural portions of the housing that are in or proximate the compression chamber, the respective plurality of bearing chambers being configured to securely position the respective bearing packs, the plurality of bearing chambers including a forward bearing chamber and an aft bearing chamber for each of the rotors, the forward and aft bearing chambers being fluidly connected with each other through the compression chamber of each of the rotors; the fluid is directed to the forward and aft bearing chambers for each of the rotors within the compressor that are fluidly coupled to the compression chamber; and upon activation of the compressor, the fluid is injected to one side of the plurality of bearing chambers and flows through the plurality of bearing chambers to lubricate each of the plurality of bearing packs and is distributed about the rotors; and the condenser conduit includes a forward branch and an aft branch for injecting in parallel the working fluid to the forward bearing chamber and the aft bearing chamber for each of the rotors of the compressor, each of the forward and aft branches of the condenser conduit includes a plurality of sub-branches for injecting in parallel the working fluid to the plurality of bearing chambers on each of the forward and aft branches of the condenser conduit.

12. The method of claim 11, comprising: controlling flow through the first port with a flow control orifice.

13. The method of claim 11, wherein: directing the fluid to the compression chamber includes: injecting the fluid from the first port directly into the compression chamber.

14. The method of claim 11, wherein: directing the fluid to the compression chamber includes: injecting the fluid from the first port, through a passage in one rotor of the pair of rotors, whereby the fluid is injected into the compression chamber.

15. The method of claim 14, wherein: injecting the fluid through the passage includes: directing the fluid from the first port into an axial aft port in the passage and out an outer surface of the one rotor.

16. The method of claim 15, wherein: directing the fluid through the passage further includes: directing the fluid through an axial segment forming a blind hole in the one rotor and a radial segment fluidly connected between the axial segment and a first surface port on the outer surface of the one rotor.

17. The method of claim 16, wherein: directing the fluid through the passage further includes: directing the fluid through a plurality of the radial segments fluidly connected to a respective plurality of the surface ports on the outer surface of the one rotor.

18. The method of claim 17, wherein: the plurality of surface ports are staggered at regular intervals along the outer surface of the one rotor.

19. The method of claim 18, wherein: directing the fluid through the passage further includes:

directing the fluid through opposing radial portions of each of the plurality of the radial segments, the opposing radial portions extending to a respective plurality of the surface ports on the outer surface of the one rotor.

20. The method of claim 11, comprising: receiving the fluid at the first port from a condenser in a refrigerant system in which the compressor is integrated, to provide the working fluid to each rotor.
